Evaluating NBA Contract Efficiency Through Performance-Based Value and Return on Investment Framework

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Abstract

This project uses data analytics to evaluate the value of NBA players relative to their salaries for a single season. The process includes collecting and cleaning player performance and salary data, calculating a comprehensive performance statistic, and computing each player's return on investment relative to their performance. The analysis identifies the most and least cost-effective players and examines team-level ROI to determine which franchises are getting the most value from their roster. The study highlights patterns in player and team value, demonstrating how data-driven analysis can provide meaningful insights in professional sports.

1 Introduction

Teams in the National Basketball Association (NBA) have to operate under a hard salary cap that creates financial restrictions on roster building. Teams are allowed to exceed the salary cap but in doing so they incur penalties that limit team-building flexibility. Penalties include a luxury tax, apron system, and roster restrictions N. B. P. A. . N. B. Association (2023). Due to this salary cap, the ability for front offices to maximize every dollar spent is vital to an NBA team's operations. Inefficient player contracts can consume cap space that could be otherwise spent on more productive contributors and can negatively impact a team's championship aspirations. In order to better navigate these struggles, more and more front offices are using quantitative approaches in order to analyze players and their contract situation Wang, Sarker, and Hosoi (2025). Some of these quantitative approaches include Win Shares, Player Efficiency Rating, Real Plus-Minus and other player valuation methods. This paper presents a framework adapted from the variable set and weighting coefficients established by Lautier (2025), simplifying certain components of their model to prioritize interpretability and direct applicability to contract evaluation. Using data from the 2024-2025 NBA regular season, the framework examines player on-court performances to see which ones generated value exceeding their contract cost and which ones under performed making them not worth what they are being paid.

2 Background

Evaluating NBA players in financial terms has become increasingly important as front offices try to navigate the salary cap. Teams must justify large maximum contracts not just by performance, but also in comparison to their internal roster and the broader player market. Traditional player valuation methods such as Player Efficiency Rating (PER), Win Shares, and Real Plus-Minus (RPM) each capture important aspects of player performance. PER evaluates a player's per-minute efficiency, Win Shares examines a player's contribution to team victories, and RPM estimates individual player impact through regression analysis. While these metrics are useful for evaluating

performance, they do not express player value in financial terms, leaving front offices to bridge the gap between performance and salary themselves.

Lautier (2025) address this need with a return on investment (ROI) framework that directly compares player performance to salary. Their approach uses logistic regression to identify which in-game actions — shooting, rebounding, defense, and hustle — contribute most to winning. These win probabilities are then converted into cash flows and used to calculate an internal rate of return against a player's salary. The framework also applies variable game weights to account for contextual differences such as home versus away games, playoff implications, opponent strength, and nightly roster composition. Together, these elements capture both the quality of a player's performance and the financial implications of each contest.

While this paper's core variable set and performance weights are drawn from Lautier (2025), the ROI calculation process is simplified in two ways. First, Lautier et al. weight each game differently based on contextual factors such as location, and game stakes. This paper instead weights all games equally, assigning each a dollar value derived from the total salary pool divided by the total number of games in a season. Second, Lautier et al. use logistic regression to convert player performance scores into win probabilities, which are then used to solve for an internal rate of return. This framework replaces that approach with a linear performance score that is converted into a fractional share of a game's value and multiplied by the uniform game value, producing a per-game cash flow that is then compared to a player's annual salary to arrive at a final ROI figure. This linear approach sacrifices some of the nuance captured by win probabilities, but the tradeoff is a more interpretable metric that does not require familiarity with internal rate of return concepts or nonlinear optimization. Ultimately, this framework is intended to serve as a practical financial evaluation tool for front offices, enabling direct comparison of a player's on-court production to their salary.

3 Data Collection

The data for this paper comes from two main sources. The first is the nba api Python package, which provides all player performance data. Specifically, two datasets were obtained to gather the variables needed for the 24 metrics in the framework. The first is the game-by-game box score logs produced from every NBA game, used for basic box score variables such as points, rebounds, assists, and shooting splits. The second is season-level advanced tracking statistics, which include rebounding tracking, defensive tracking, hustle stats, passing data, and speed/distance metrics. All performance data was drawn from the 2024–25 NBA regular season.

The second source is ESPN's salary tracking page, from which player salary data was obtained via web scraping using Python's BeautifulSoup library. The scraped data includes each player's name, team, and cap hit for the 2024–25 season. It is worth noting that cap hit reflects the base salary charged against a team's payroll and excludes performance bonuses.

4 Data Cleaning

With data coming from two different sources, there was some inconsistency between them that needed to be resolved before merging. The first issue was player name formatting. In the salary table, each player's name was followed by a comma and a one-letter position abbreviation — both of which needed to be stripped out. Beyond that, any commas, periods, dashes, or other special characters within names were removed, and everything was converted to uppercase so the two datasets could eventually be joined on a common key. The salary values themselves also needed attention. Dollar signs and commas were stripped, and the resulting strings were converted to integers so they could be used in calculations later. An issue came up with traded players — some appeared twice in the salary data because they had been dealt mid-season. Since the salary figures were the same in both rows, the duplicates were simply dropped, keeping only one record per player.

With the data clean, some feature engineering was done to match the structure required by the

Lautier et al. framework. The framework calls for a specific breakdown of 2-point and 3-point shooting, so new columns were created accordingly. It also requires separating made shots from misses rather than working with attempts and makes, so each new column was given a suffix — O for made (scored) and X for missed — to keep things clear. Finally, the season-level advanced tracking stats were merged onto the per-game box score logs using each player's ID. For any player without available tracking data, zeros were filled in.

5 ROI Framework

The ROI framework is built around five steps to evaluate a players performance relative to their salary.

5.1 Performance Score

The performance score measures how well a player performed compared to the league average on a game-by-game basis. The formula is as follows:

$$\textit{Performance Score} = \mathcal{E}(\textit{Weight}(\textit{Players Stat-League Average Stat}))$$

The statistics are centered so that a perfectly league-average game yields a performance score of zero. The 24 variables and their corresponding weights, taken from Lautier (2025), are presented in the tables below.

In Table 1, the variables carrying the strongest positive and negative weights are 3-point field goals made (+0.537) and 3-point field goals missed (-0.368), respectively. This makes intuitive sense — three-point shooting is the highest-leverage offensive action in the modern NBA, and the framework reflects that in both directions. Converting a three-pointer is worth more than any other shot a player can take, hence the highest positive weight in the table. Conversely, missing a three-pointer is particularly costly because it not only wastes a possession but also tends to produce

Variable	Abbreviation	Weight
2-point field goals made	FG2O	0.251
2-point field goals missed	FG2X	-0.349
3-point field goals made	**FG3O**	**0.537**
3-point field goals missed	**FG3X**	**-0.368**
Free throws made	FTO	0.122
Free throws missed	FTX	-0.220
Personal fouls committed	PF	-0.197
Steals	STL	0.443
Blocks	BLK	0.132
Turnovers	TO	-0.347
Personal fouls drawn	PFD	0.214

Table 1: Basic Box Score Statistics along with their Weights

Variable	Abbreviation	Weight
Uncontested offensive rebound	AORB	0.356
Uncontested defensive rebound	ADRB	0.316
Contested offensive rebound	OCRB	0.290
Contested defensive rebound	DCRB	0.338
Defended field goal made	DFGO	-0.230
Defended field goal missed	DFGX	0.086
2-point field goal contetsted on defense	AC2P	0.041
3-point field contested on defense	C3P	-0.067
Screen assist	SAST	0.076
Charge drawn	**CHGD**	**0.522**
Defensive box out	DBOX	0.053
Defensive distance traveled	**DDIS**	**-1**
Non assist passes made	APM	0.016

Table 2: Advanced Tracking Statistics along with their Weight

long rebounds that are harder for the offense to recover, giving the defense a favorable transition opportunity.

In Table 2, the variables with the strongest positive and negative weights are charges drawn (+0.522) and defensive distance traveled (-1.000). Charges drawn carry a high positive weight because they are one of the most impactful defensive plays a player can make — drawing a charge not only forces a turnover but also gives the ball handler a personal foul, two outcomes that can swing momentum in a single moment. Defensive distance traveled is a more nuanced case, though, and could honestly be interpreted either way. A player covering more ground on defense might simply be working harder and protecting more of the court, but it could just as easily reflect poor positioning. The framework assigns it a negative weight under the assumption that high defensive distance is more often a sign of defensive strain than defensive impact. This is revisited in the limitations section, as there are questions about whether that interpretation is always the case.

5.2 Player Value Share

A player's value share captures how much of a game's total value belongs to each player based on their performance score. The calculation breaks down into three steps.

Step 1 - Calculate the players Z-score:

$$\textit{Player Z-score} = \frac{\textit{PS} - \textit{NBA Mean PS}}{\textit{NBA PS STD}}$$

Step 2 - Calculate the replacement share for a game:

$$\textit{Game Share} = \frac{1}{\textit{Mean Number of Players in a Game}}$$

Step 3 - Calculate the players value share for a single game:

$$\textit{Player Value Share} = \textit{Z-score} \times \textit{Game Shares}$$

Average level players earn exactly $\frac{1}{\textit{Mean Number of Players in a Game}}$ - the replacement level share. Better-than-average performances earn more than that, while below-average performances earn less.

5.3 Single Game Value

The Single Game Value (SGV) takes a uniform approach, dividing the league's total salary pool across all 30 teams and 82 games per season. The formula is straightforward:

$$SGV = \frac{\textit{Total Salary Pool}}{\textit{(Total Teams} \times \textit{Games in a Season}}$$

Based on the salary data collected for this study, the total 2024–25 salary pool came out to \$5,169,349,665 spread across 2,460 team-games (30 teams × 82 games), the actual dollar value per game came out to be \$2,101,361.65.

5.4 Per Game Cash Flow

Using the previously calculated Player Value Share and Single Game Value (SGV), we can determine how much a player "earned" for any given game based on their performance. We call this their Cash Flow for that game, calculated as:

$$\textit{Cash Flow} = SGV \times \textit{Player Value Share}$$

One important thing to note: if a player is unable to play in a game, their Cash Flow for that game is simply zero. Since they didn't contribute anything on the court, they hold no share of that game's value. This naturally penalizes availability — a player who misses time accumulates less total Cash Flow over the course of a season, regardless of how well they perform when healthy.

5.5 Final ROI Value

With all the pieces in place, we can finally compute a single number that captures whether a player lived up to their contract. The formula is straightforward:

$$ROI = Total\ Season\ Cash\ Flow \div Salary$$

A value of 1.0 is the break-even point — it means a player generated exactly as much value as they were paid. Anything above 1.0 means they outperformed their contract, and anything below 1.0 means they fell short of justifying their salary.

6 Results

Figure 1 presents a scatter plot of every player's ROI against their salary, color-coded by contract tier. Minimum-contract players cluster near the top of the ROI axis, while the highest-paid players sit near or below the break-even line of 1.0. Across the full player pool, minimum-contract players averaged an ROI of 2.78, compared to 0.69 for mid-level earners, 0.26 for starter-level contracts, and 0.14 for supermax deals. This reflects a structural dynamic in NBA roster construction — teams are paying a premium for established stars while getting surplus value from younger, cheaper players who are still performing near or above replacement level.

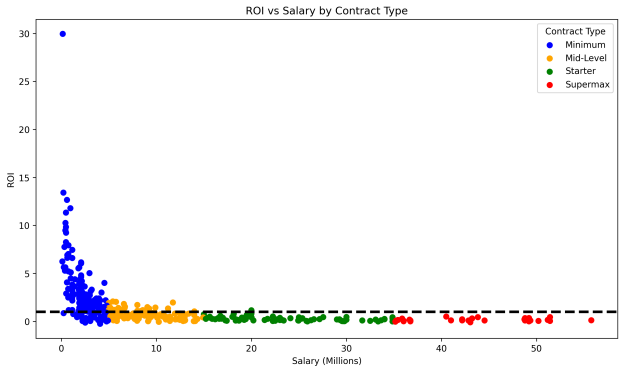


Figure 1: Player ROI vs Player Salary

To examine this more closely among players with meaningful playing time, the analysis was restricted to those who met the NBA's 65-game eligibility threshold for end-of-season awards such as MVP and Defensive Player of the Year N. B. P. A. . N. B. Association (2023). Of the 169 players who qualified, 71 (42%) exceeded the break-even ROI of 1.0. The mean ROI in this group was 1.301, while the median was 0.767 — a gap that signals right skew driven by a handful of minimum-contract players who dramatically outperformed their deals.

Figure 2 illustrates this disparity directly, showing the top 10 and bottom 10 ROI players among those who qualified under the 65-game rule. The top performers are almost entirely players on minimum or near-minimum contracts. At the other end, the lowest ROI values belong largely to high-salary players: Keyonte George, Isaiah Collier, and Trae Young all posted negative ROIs, meaning the model estimates their on-court production did not justify their contract cost in the 2024–25 season.

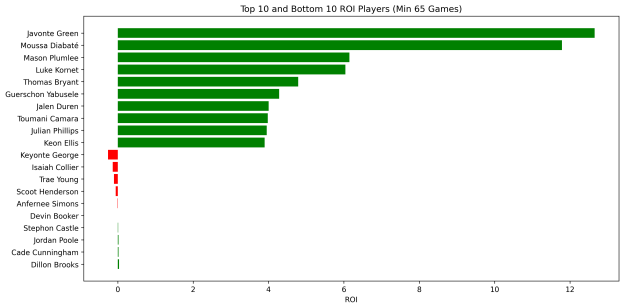


Figure 2: Top 10 and Bottom 10 ROI Players (65-game rule)

Figure 3 plots each team's average player ROI against its win percentage for the 2024–25 season. The Pearson correlation between the two variables is r = 0.17, suggesting only a weak positive relationship — teams with higher payroll efficiency did not reliably win more games. The chart is divided into four quadrants by the median ROI and the .500 win threshold, and the distribution across those quadrants tells an interesting story. The Cleveland Cavaliers and Boston Celtics sit in the efficient-and-winning quadrant, suggesting their roster construction translated both on the court and in value terms. The Oklahoma City Thunder are a notable outlier in the opposite direction — the best record in the league yet landing in the inefficient-and-winning quadrant, driven by

the large contracts of their young stars which the model penalizes relative to their output. On the other end, teams like the Sacramento Kings and Golden State Warriors posted high average ROIs despite finishing below .500, reflecting rosters padded with cheap contributors but lacking the star power to win consistently. The weak overall correlation suggests that raw payroll efficiency, as measured here, is not a strong standalone predictor of team success — winning in the NBA still depends heavily on having elite players, even if those players are expensive relative to what the model values.

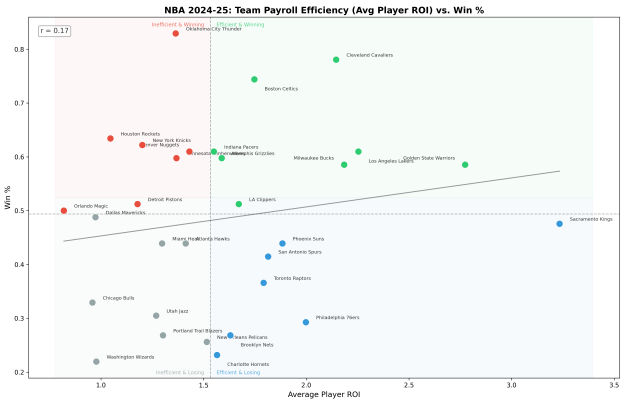


Figure 3: AVG Team ROI vs Win Percentage

7 Takeaways

The main takeaway of this study is that minimum or near-minimum contract players are where teams are going to find the most efficient value in the NBA. Front offices that can identify these high-value, low-salary players are the ones that will help their teams win while preserving roster flexibility. Meanwhile, starter and supermax-level contracts rarely justify their price tags under this framework.

Availability also matters enormously — if a player misses games, their team gets zero value from them. The players generating strong ROI are the ones who stay healthy and can produce on a game-by-game basis.

Defense is also notably underpriced by the NBA market, though not by this framework. Statis-

tics like steals and charges drawn carry very high weights here but are rarely reflected in salaries. Players with good ROI tend to excel in these areas.

From a practical standpoint, the key lessons for front offices are to prioritize availability, seek out defensively active players on minimum or near-minimum contracts, and use those savings to build cap flexibility around one or two max-level cornerstones.

8 Limitations and Future Improvements

The clearest finding from this study is that minimum and near-minimum contract players represent the most efficient value in the NBA. Front offices that can identify these high-value, low-salary contributors stand to gain both on the court and in terms of long-term roster flexibility. Starter and supermax-level contracts, by contrast, rarely justify their price tags under this framework — which makes sense when you think about it. The market prices star power well above what any single player can realistically return in measurable on-court value, and that gap shows up clearly in the numbers.

Availability is just as important as performance. A player who misses games contributes nothing to their team for those nights, and the framework captures this directly — missed games produce zero cash flow. The players generating the strongest ROI are almost always the ones who stay healthy and show up consistently, which is a reminder that durability is one of the most underrated qualities a roster-builder can look for.

Defense also stands out as an underpriced asset in the broader NBA market. Statistics like steals and charges drawn carry some of the highest positive weights in the model, yet based on the results here, the players who excelled in these areas were rarely among the highest-paid on their rosters — suggesting the market may not be fully pricing in defensive production.. Teams that seek out defensively active players on modest contracts — and use the savings to build flexibility around one or two max-level cornerstones — are probably the ones best positioned to get real value out of a framework like this.

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